Your grade: 92.33%

Your latest: **92.33%** • Your highest: **92.33%** • To pass you need at least 80%. We keep your highest score.

Full Course Assessment

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Due
Dec 16, 11:59 PM IST

Attempts
Unlimited

Retry

Submitted
Dec 9, 11:19 PM IST

Your grade
To pass you need at least 80%. We keep your highest score.

92.33%

View submission See feedback

Like Dislike Report an issue

---

1. What is the impact of quantum computing on solving complex problems compared to classical computing? **1 / 1 point**

○ Quantum computing is slower than classical computing for all types of problems.
● Quantum computing can solve certain complex problems exponentially faster than classical computing.
○ Quantum computing solves problems at the same speed as classical computing.
○ Quantum computing has no impact on problem-solving speeds.

✓ **Correct**
Correct! Quantum computing uses quantum bits (qubits) which can represent multiple states simultaneously, significantly speeding up computations.

2. What is the result of dividing the complex number **0 / 1 point**

$$(6+2i)$$

by

$$(3-i)$$

in quantum computations?

○ $3-2i$

○ $2-i$

● $1+3i$

○ $2+i$

✗ **Incorrect**
Incorrect. Ensure you are using the complex conjugate correctly in your calculations.

3. In the context of quantum computing, what is the main significance of the identity matrix? **1 / 1 point**

○ It represents a matrix with all entries equal to zero.
● It acts as the neutral element in matrix multiplication, meaning any matrix multiplied by it remains unchanged.
○ It is used to calculate the determinant of other matrices.
○ It is only applicable in classical computing, not in quantum computing.

✓ **Correct**
Correct! The identity matrix is the neutral element in matrix multiplication.

4. Which of the following are fundamental differences between classical and quantum computers? **1 / 1 point**

☑ Quantum computers can process multiple states simultaneously through superposition.

✓ **Correct**
Correct! Superposition allows quantum computers to process multiple states at once, unlike classical computers.

☐ Classical computers use probabilistic algorithms while quantum computers use deterministic algorithms.
☑ Quantum computers use qubits while classical computers use bits.

✓ **Correct**
Correct! This is one of the primary differences between classical and quantum computers.

☐ Classical computers are capable of factoring large numbers more efficiently than quantum computers.
☑ Quantum computers rely on quantum entanglement for processing.

✓ **Correct**
Correct! Quantum entanglement is a key feature in quantum computing, enhancing computational power.

5. What is the significance of eigenvalues and eigenvectors in quantum computations? **1 / 1 point**

○ They are used in the calculation of tensor products.
● They help in determining the probabilities of different quantum states.
○ They are used to add and subtract complex numbers.
○ They help in performing scalar multiplication of matrices.

✓ **Correct**
Correct! Eigenvalues and eigenvectors play a crucial role in understanding quantum states and their probabilities.

6. When adding two complex numbers, **1 / 1 point**

$$(3+2i)$$

and

$$(1-4i)$$

, in the context of quantum computing, what is the result?

○ $(2-6i)$

● $(4-2i)$

○ $(4-6i)$

○ $(3-2i)$

✓ **Correct**
Correct! Remember to add the real parts and the imaginary parts separately.

7. Which property is unique to unitary matrices in quantum computations? **1 / 1 point**

○ They are always symmetric
● They preserve the inner product of vectors
○ Their determinant is always zero
○ They have only real eigenvalues

✓ **Correct**
Correct! Unitary matrices preserve the inner product of vectors, which is essential in quantum computing.

8. When performing matrix multiplication in quantum computing, what is the requirement for the dimensions of the two matrices? **1 / 1 point**

○ The number of rows in the first matrix must equal the number of columns in the second matrix.
○ Both matrices must have the same number of columns.
○ Both matrices must have the same number of rows.
● The number of columns in the first matrix must equal the number of rows in the second matrix.

✓ **Correct**
Correct! This is a fundamental requirement for matrix multiplication.

9. Which of the following operations are characteristics of classical computers but not quantum computers? **1 / 1 point**

☑ Data representation using binary bits

✓ **Correct**
Correct! Classical computers use binary bits for data representation.

☐ Simultaneous evaluation of multiple states
☐ Data processing using superposition
☑ Operations based on deterministic algorithms

✓ **Correct**
Correct! Classical computers often rely on deterministic algorithms.

10. Which of the following are properties of unitary matrices in the context of quantum computing? **0.8 / 1 point**

☐ The matrix is equal to its conjugate transpose.
☐ The matrix is equal to its inverse.
☑ All eigenvalues have an absolute value of 1.

✓ **Correct**
Correct! Unitary matrices have eigenvalues with an absolute value of 1.

☐ All entries in a unitary matrix are real numbers.
☐ The determinant of a unitary matrix is always 1.

You didn't select all the correct answers

11. Which concepts are fundamental to understanding probability in quantum computing? **1 / 1 point**

☐ Matrix inversion
☐ Transcendental numbers
☑ Mutually exclusive events

✓ **Correct**
Correct! Mutually exclusive events are fundamental in probability theory.

☑ Independent events

✓ **Correct**
Correct! Independent events are also fundamental in understanding probability.

12. What is the importance of the complex conjugate in the division of complex numbers in quantum computations? **1 / 1 point**

● It helps in rationalizing the denominator
○ It simplifies the division to obtain a real number result
○ It is used to eliminate the imaginary part
○ It converts the complex number to its polar form

✓ **Correct**
Correct! The complex conjugate is used to rationalize the denominator during division.

13. What principle of quantum mechanics explains why measuring one particle of an entangled pair immediately determines the state of the other particle? **1 / 1 point**

○ Quantum interference
○ Wave function collapse
○ Quantum superposition
● Quantum entanglement

✓ **Correct**
Correct! Quantum entanglement is the phenomenon where the quantum states of two or more objects are interconnected, such that the state of one instantly influences the state of the other, no matter the distance separating them.

14. Which of the following principles are essential for understanding quantum computing? **0.6666666666666666 / 1 point**

☐ Error correction
☑ Entanglement

✓ **Correct**
Correct! Entanglement enables qubits to be interconnected, which is crucial for quantum algorithms and protocols.

☐ Classical bits
☑ Superposition

✓ **Correct**
Correct! Superposition allows qubits to exist in multiple states simultaneously, a fundamental principle of quantum computing.

☑ Wave function collapse

✗ **This should not be selected**
Incorrect. While wave function collapse is a quantum mechanics concept, it is not a principle that enhances computational power in quantum computing.

☑ Interference

✓ **Correct**
Correct! Interference is used in quantum algorithms to amplify correct solutions and cancel out incorrect ones.

15. In quantum computation, why is the concept of superposition important? **1 / 1 point**

○ It provides a way to measure quantum interference patterns.
● It allows qubits to represent both 0 and 1 simultaneously, increasing computational power exponentially.
○ It ensures that quantum entanglement can occur between qubits.
○ It prevents quantum states from collapsing during computation.

✓ **Correct**
Correct! Superposition enables qubits to perform many calculations at once, significantly enhancing the computational capabilities of quantum computers.

16. Which statements are true about the representation of qubit states in quantum mechanics? **1 / 1 point**

☑ Qubit states can be in a superposition of 0 and 1

✓ **Correct**
Correct! Qubit states can exist in a superposition of both 0 and 1 simultaneously.

☑ Qubit states can be represented using column vectors

✓ **Correct**
Correct! Qubit states are often represented using column vectors in quantum mechanics.

☐ Qubit states are always in a definite state of 0 or 1
☐ Qubit states are represented as classical bits
☐ Qubit states cannot be entangled

17. Which of the following are true about entanglement in quantum mechanics? **1 / 1 point**

☐ Entanglement is a classical phenomenon.
☑ Entanglement can be described using Bra-Ket notation.

✓ **Correct**
Correct! Entanglement can be mathematically described using Bra-Ket notation.

☑ Entangled particles have correlated properties.

✓ **Correct**
Correct! Entangled particles exhibit correlated properties regardless of the distance separating them.

☑ Measurement of one entangled particle determines the state of the other.

✓ **Correct**
Correct! Measuring one entangled particle immediately determines the state of the other particle.

☐ Entanglement only occurs at very low temperatures.

18. Which of the following best describes the concept of interference in quantum mechanics? **1 / 1 point**

○ The representation of quantum states using Bra-Ket notation.
○ The collapse of the wave function upon measurement.
○ The process by which quantum states are entangled with each other.
● The phenomenon where multiple quantum states add together and the probability amplitudes combine.

✓ **Correct**
Correct! Interference is a fundamental concept in quantum mechanics where the probability amplitudes of different quantum states combine.

19. How is the tensor product used in quantum computing? **1 / 1 point**

○ To normalize quantum states.
○ To measure the entanglement between qubits.
● To combine multiple quantum states into a single state.
○ To represent the probability of measurement outcomes.

✓ **Correct**
Correct! The tensor product combines multiple quantum states into one.

20. What is the significance of measurement alignment in quantum spin measurement? **1 / 1 point**

○ It influences the speed of quantum computation.
● It determines the probability distribution of the measured spin state.
○ It changes the entanglement between particles.
○ It collapses the superposition state to a classical state.

✓ **Correct**
Correct! Measurement alignment affects the probability distribution of the measured spin state.